

THE ELEMENTS OF MICROECONOMICS

*Prices, Markets, Costs and Rewards
explained from first principles*

A PRIMER FOR FIRST-YEAR STUDENTS

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Prices, Markets, Costs and Rewards, explained from first principles

A Primer for First-Year Students of Business Studies

British spelling. Currency: the Bangladeshi taka (Tk). The bakery used in the examples is *illustrative*: it is not a real business. All figures in the examples are invented to teach method.

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TO THE READER

This book teaches **microeconomics**, the study of how people, households and firms make choices and how those choices meet in markets. You need no earlier knowledge of business. You need only arithmetic and basic algebra.

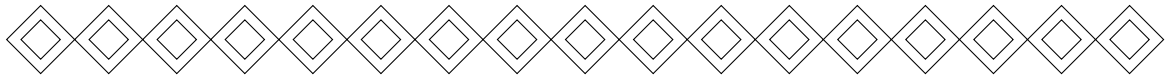
How a page is built. The wide column holds the main text. The narrow column at the outer edge holds notes. An *italic* note gives the one-line point of the paragraph beside it. A note that begins with a word in **bold type** gives the plain meaning of a key term.

Bold type in the main text marks a key term at its first use only. Italics are used for emphasis, for titles of books and for foreign words. Each key term is explained in simple English, then shown in an example. A full list appears at the back, in the Glossary and Index, with the page of first use.

How a chapter is built. Each chapter opens with a question. Each idea then follows four steps: the idea, an example, the rule, and the exceptions. A **worked example** shows every step of a calculation in the same order: Given, Find, Formula, Steps, Answer, Check. A **case file** follows one small bakery through the book. The bakery is invented for teaching.

How to study. Read slowly. Where a passage ends with “You should now be able to say”, test yourself aloud before you move on. Do the review questions at the end of each chapter, then check your answers at the back.

The mathematics. Percentage change means the change divided by the starting value, multiplied by 100. A ratio compares two numbers by division. To read a graph, find a point on the horizontal axis, go up to the curve, then read across to the vertical axis. Nothing else is needed.



CHAPTER THE FIRST

WHAT ECONOMICS IS ABOUT

Why can nobody have everything?

The chapter begins with one oven and two choices. It shows why choice is unavoidable, defines economics and its two branches, and explains what an economic law is.

On 1 January 2024 the owner of a small bakery in Dhaka opens for the first time. She has one oven and eight hours a day to use it. In one day the oven can bake 400 loaves of bread or 80 cakes. It cannot bake both at full strength. Which should she bake? Any answer gives something up. Baking cakes means fewer loaves, and baking loaves means no cakes. This small problem is the whole of economics in miniature.

Start with a question you can almost answer.

CASE FILE: RUPALI BAKERY (ILLUSTRATIVE)

1 Jan 2024: opens in Mirpur, Dhaka. One oven, eight oven hours a day. Output choice per day: 400 loaves or 80 cakes, or a mixture in the same ratio (1 cake = 5 loaves). The owner is the only decision-maker.

THE MEANING OF ECONOMICS

Economics means the study of how people choose to use limited things to satisfy their wants. A **want** is anything a person would like to have, such as food, a phone or a holiday. A **resource** is anything used to make goods and services, such as land, labour, machines and money. The baker's oven hours are a resource. Her customers' wish for bread and cake is a want.

The central fact is **scarcity**. Scarcity means that resources are limited while wants are without end. Scarcity does not mean poverty. A rich country and a poor country both face it. Even a person with great wealth has only twenty-four hours in a day. Because of scarcity, every choice has a price, and that price is what we give up.

Writers have defined economics in several ways. Adam Smith (1776) described it as an inquiry into the nature and causes of the wealth of nations. Alfred Marshall (1890) called it a study of mankind in the ordinary business of life, and linked it to human welfare. Lionel Robbins (1932) wrote that it studies human behaviour as a relation between ends and scarce means that have alternative uses. Most modern teachers prefer the last, because it applies to every choice, not only to money.

The three definitions differ in their focus. Smith's focuses on wealth, Marshall's on welfare, and Robbins's on scarcity and choice. Critics said Smith

Economics the study of how people choose to use limited resources to satisfy wants.

The idea: choice under limits.

Scarcity resources are limited while wants are not.

Scarcity is the root of all economic problems.

Three famous definitions.

Each definition looked at a different part.

Opportunity cost the value of the best alternative given up when a choice is made.

The real cost of a choice is what is given up.

ignored well-being, and that Marshall's definition left out services that do not add to welfare. Robbins's definition covers both, but it says little about growth over time. For a first course, remember this: economics is about choice made necessary by scarcity.

The baker's problem shows a key idea. If she bakes 80 cakes, she gives up 400 loaves. The loaves given up are the **opportunity cost** of the cakes. Opportunity cost means the value of the best alternative that you give up when you choose. We study it fully in Chapter 9. For now, hold the idea: choosing one thing means not choosing another.

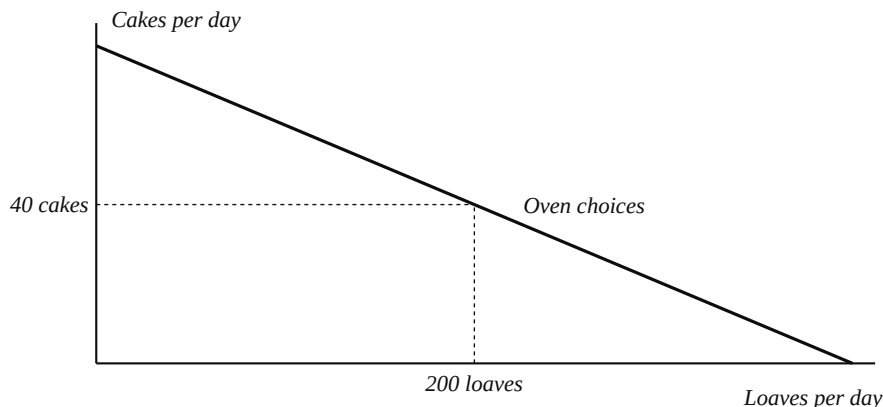


FIG. 1. Every point on the line is one possible day's baking. The point shown is a mixture.

Reading the figure.

Read the figure in three steps. The horizontal axis counts loaves, and the vertical axis counts cakes. The line joins the two extreme choices: all cakes at the top, all loaves at the right. A point on the line, such as 200 loaves and 40 cakes, shows a mixture. To gain more cakes, you must move left, and that means fewer loaves.

You should now be able to say: scarcity forces choice, and every choice gives something up.

You should now be able to say: "Resources are scarce, so a choice for one thing is a choice against another." Test yourself before you read on.

THE BASIC PROBLEMS OF AN ECONOMY

Economic system the way a society organises the choices of what, how and for whom to produce.

Every society answers three questions.

Two further problems follow from these three.

Every society, rich or poor, must answer three questions. **What** goods shall be produced, and in what amounts? **How** shall they be produced, by hand or by machine, in small shops or large factories? **For whom** shall they be produced, that is, how shall the output be shared? The baker answers all three for her shop. A country answers them for millions of people, through its **economic system**.

Two more problems follow. Society must use its resources fully and without waste, and it must decide how much to save for the future so that output can grow. Unused ovens and idle workers are a waste. Both problems still rest on scarcity. Markets, governments or a mixture of both give the answers. Bangladesh is a mixed economy: private firms and the state both take part.

MICRO AND MACRO

Microeconomics the study of the choices of single households, firms and markets.

Micro looks at the parts.

Macroeconomics the study of the whole economy, such as total output, prices and jobs.

Macro looks at the whole.

Microeconomics means the study of the choices made by single households and firms, and of the single markets where they meet. The word comes from the Greek *mikros*, "small". It asks why a loaf costs Tk 30, why a baker hires four workers, or why bread sells more than cake.

Macroeconomics means the study of the whole economy: total output, the general level of prices, employment and the money supply. The word comes

from *makros*, “large”. It asks why prices rise across the country, or why total output grows slowly.

MICRO AND MACRO COMPARED

Point of difference	Microeconomics	Macroeconomics
Unit studied	One household, firm or market	The whole economy
Main question	How is price set?	What decides total output?
Main tools	Demand, supply, cost	Income, spending, money
Example	The price of bread	The rate of inflation

The two branches are not rivals. Total output is the sum of many single choices, and each firm’s choices depend on the state of the whole economy. A fall in national income reduces bread sales. Still, you must learn the parts before the whole, and that is why this book starts with micro.

The two are linked.

THE SCOPE AND USE OF MICROECONOMICS

The **scope** of a subject means the range of topics it covers. Microeconomics covers the theory of demand, which explains buyers; the theory of supply and production, which explains sellers; the theory of the firm and of market structures, which explains competition and monopoly; and the theory of distribution, which explains rent, wages, interest and profit. Welfare economics, which asks whether the result is good for society, is a further part.

Scope means what the subject covers.

Its objectives are practical. It explains how prices are set, so that firms can decide what to produce and at what price. It shows how resources are shared among uses. It helps governments judge taxes, subsidies and price controls. It also helps a consumer plan spending. A firm that knows its costs and its customers decides better.

Why the study is useful.

In Bangladesh the subject has particular value. The country has many small farms, workshops and shops, so market prices matter to millions of people. Planners need to judge the effect of taxes and price rules on a developing economy. Support for firms comes from institutions such as Bangladesh Bank, the Bangladesh Small and Cottage Industries Corporation (BSCIC) and the Bangladesh Competition Commission. These bodies rely on microeconomic ideas when they set rules for credit, small enterprise and fair competition.

Prospects in Bangladesh.

ECONOMIC LAWS, SCIENCE AND ART

An **economic law** means a statement of how people usually behave in economic matters. An example: when the price of a good rises, buyers tend to buy less. Such laws differ from the laws of physics in four ways. They state tendencies, not certainties, because human beings choose. They hold only if other things stay the same, a condition known by the Latin phrase *ceteris paribus*. They are less exact, as they cannot be tested in a laboratory. And they apply to groups of people more than to one person.

Economic law a statement of the usual relation between economic events, true when other things stay equal.

A law states a tendency, not a certainty.

Economics contains two kinds of statement. A **positive** statement says what is, and facts can test it: “A higher bread price reduces bread sales.” A **normative** statement says what ought to be, and it rests on values: “The government should fix the price of bread.” Positive economics explains. Normative economics judges. Good reasoning keeps the two apart, because facts can settle the first but not the second.

Positive statement a statement about what is, which facts can test.

Positive: what is. Normative: what ought to be.

Is economics a science or an art?

A **science** is an organised body of knowledge that links causes to effects. An **art** is the skilful use of knowledge to reach a goal. Economics is both. It is a science because it builds laws from observation and reasons from cause to effect. It is an art because a baker, a banker or a minister applies those laws to reach a goal. Positive economics is the science, and normative economics guides the art.

CASE FILE: DECISION ON 1 JANUARY 2024

The owner chooses 300 loaves and 16 cakes for the first day. Check against the oven: 300 loaves use 6 of 8 oven hours; 16 cakes use 1.6 hours; total 7.6 hours. The remaining 0.4 hour is unused, which is waste in the language of this chapter.

SUMMARY. Scarcity forces choice. Economics studies how people choose when resources are limited. Micro studies single decision-makers and markets; macro studies the whole economy. Economic laws state tendencies that hold when other things are equal. Positive statements describe; normative statements judge.

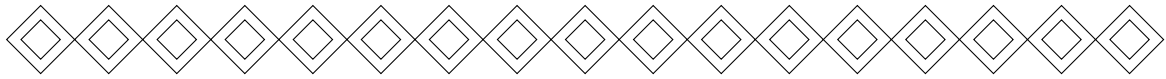
THINK IT THROUGH. A student can spend Saturday working for Tk 600 or studying for a test. She chooses to study. State what she gave up and why this is a choice caused by scarcity.

WHERE PEOPLE GO WRONG. “Scarcity means poverty.” It does not. Even the richest country cannot satisfy every want. Also, a normative opinion is not wrong because it is an opinion; it is simply not a positive statement.

IN HISTORY. In 1776 Adam Smith described a pin factory in which ten workers, each doing one task, made about 48,000 pins a day. Working alone, each could hardly make twenty. He used the case to show that dividing tasks raises output from the same scarce labour.

REVIEW QUESTIONS

1. Why does scarcity not mean poverty?
2. Give one micro and one macro question.
3. Classify: “Bread prices should be capped.” Positive or normative? Why?



CHAPTER THE SECOND

UTILITY AND SATISFACTION

Why does the fifth bun taste less sweet?

The chapter begins with a puzzle about water and diamonds. It defines utility, builds the law of diminishing marginal utility from a hungry student's buns, and shows where the law holds and where it fails.

Water is cheap, though without it we die. Diamonds are costly, though we can live without them. Why? Why does the more useful good cost less? The puzzle puzzled thinkers for a century. The answer needs a new idea: the value we place on a good depends on the extra satisfaction we gain from one more unit of it. We build that idea step by step.

Start with a question you can almost answer.

UTILITY, TOTAL AND MARGINAL

Utility means the satisfaction a person gets from using a good or service. The word does not mean usefulness in a moral sense. A cigarette has utility for a smoker, though it harms health. Utility depends on the person, the time and the place. A cold drink gives more utility on a hot day than on a cold one.

Utility the satisfaction a person gets from using a good.

Utility is satisfaction.

Two measures matter. **Total utility** (TU) means the whole satisfaction from all the units consumed. **Marginal utility** (MU) means the extra satisfaction from one more unit. In words, marginal utility equals the change in total utility divided by the change in the number of units. The word *marginal* means “at the edge”, that is, the last or the next unit.

Total utility the whole satisfaction from all the units consumed.

Marginal utility the extra satisfaction from one more unit.

Two measures: total and marginal.

WORKED EXAMPLE. Total and marginal utility

GIVEN	A hungry student eats buns. TU after 1, 2, 3, 4, 5, 6 buns is 20, 36, 48, 56, 60, 60 utils.
FIND	The marginal utility of each bun.
FORMUL	MU of the nth bun = TU(n) minus TU(n-1).
A	
STEPS	MU1 = 20 - 0 = 20. MU2 = 36 - 20 = 16. MU3 = 48 - 36 = 12. MU4 = 56 - 48 = 8. MU5 = 60 - 56 = 4. MU6 = 60 - 60 = 0.
ANSWER	The marginal utilities are 20, 16, 12, 8, 4, 0 utils.
CHECK	Add them: 20+16+12+8+4+0 = 60, the total utility. The check holds.

A **util** is an imaginary unit used to count satisfaction. No machine measures it. Is satisfaction measurable at all? Early economists, such as Jevons and Marshall, said yes: a person can state how much they would pay, and that measures it in money. This is the **cardinal** view. Later writers, such as Hicks,

A unit of measure.

said a person can only rank satisfactions as first, second and third. This is the **ordinal** view, and we use it in Chapter 5.

THE LAW OF DIMINISHING MARGINAL UTILITY

Law of diminishing marginal utility as a person consumes more units, the extra satisfaction from each further unit falls.

The more we have, the less the next one adds.

Look at the example again. The marginal utility falls with each bun: 20, 16, 12, 8, 4, 0. This is the **law of diminishing marginal utility**. It states that, as a person consumes more units of a good in a given period, the marginal utility of each extra unit falls. Total utility still rises while marginal utility is positive. It reaches its highest point when marginal utility is zero, here after the fifth bun.

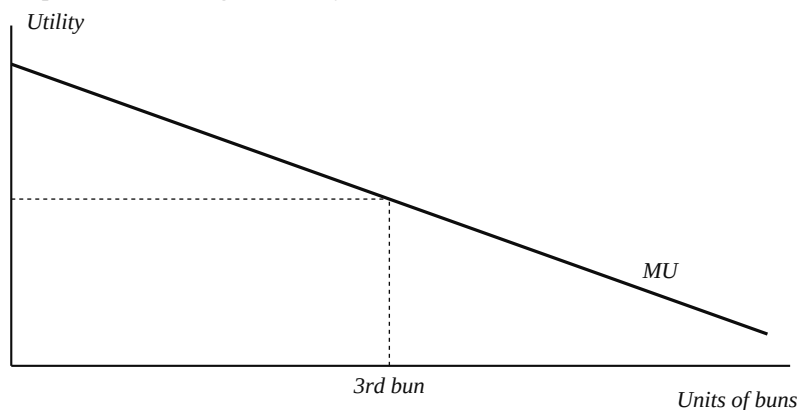


FIG. 2. Marginal utility falls as units rise (the schedule of buns, drawn smoothly).

Reading the figure.

To read the figure, find a number of buns on the horizontal axis and go up to the line. The height shows the extra satisfaction from that bun. The line slopes down, which is the law in picture form. Where the line meets the horizontal axis, the extra satisfaction is zero, and the student should stop eating.

The assumptions behind the law.

The law holds under conditions. The units must be of the same kind and of reasonable size. Consumption must be continuous, with no long breaks. The consumer's taste, income and the prices of related goods must stay the same. The consumer must act sensibly. If a condition fails, the law may appear to fail too.

Limits and exceptions.

Some cases seem to break the law. The urge for a hobby, such as collecting stamps, may grow with each stamp. A person who loves money may want more money as the pile grows. A drug user may want ever more of the drug. Knowledge may bring more interest in further knowledge. Most economists answer that in these cases a condition fails: the units are not of the same kind, the taste is changing, or the consumption is not rational. Even then, at some point the satisfaction from the next unit falls.

Back to the puzzle.

Now return to water and diamonds. Water is so plentiful that its marginal utility is low. Diamonds are rare, so the marginal utility of one more diamond is high. Price follows marginal utility, not total utility. Water has a very high total utility, but its price is low. In 1871 Jevons and Menger, and in 1874 Walras, each gave this answer separately.

CASE FILE: CUSTOMER SURVEY, 20 JAN 2024

Rupali Bakery asks one customer how much she would pay for each bun in a day. Answers: Tk 20 for the first, 16 for the second, 12 for the third, 8 for the fourth, 4 for the fifth. These are her marginal utilities in money. At a price of Tk 10 she buys three buns.

SUMMARY. Utility is satisfaction. Total utility is the whole; marginal utility is the extra from one more unit. As consumption rises, marginal utility falls. The law needs stated conditions, and the apparent exceptions break those conditions. Value follows marginal utility.

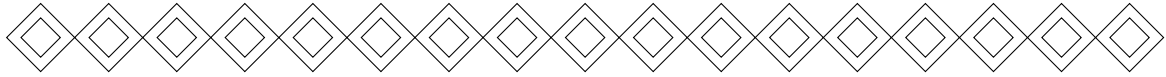
THINK IT THROUGH. Using the survey, how many buns would the customer buy at Tk 5? Explain by comparing price with marginal utility.

WHERE PEOPLE GO WRONG. “Diminishing utility means total utility falls.” It does not. Total utility rises more slowly, and it falls only when marginal utility becomes negative.

IN HISTORY. In 1738 Daniel Bernoulli argued that an extra ducat means less to a rich man than to a poor man. He used it to explain risky choices. The marginal revolution of 1871-74 turned this insight into the basis of value theory.

REVIEW QUESTIONS

1. Define marginal utility in one sentence.
2. If TU is 30, 50, 60, 62, what is MU of the third unit?
3. Name two conditions the law needs.



CHAPTER THE THIRD

THE THEORY OF DEMAND

Why do we buy less when prices rise?

The chapter begins with a bakery price change. It defines demand, states the law, builds the demand curve from a schedule, and examines the exceptions and the forces that shift the curve.

On 1 March 2024 the bakery raises the price of a loaf from Tk 25 to Tk 30. By the end of the week, fewer loaves are sold. Was this a surprise? Why does a higher price so often meet lower sales? The answer is the oldest regularity in economics, and it is the first half of the story of every market.

Start with a question you can almost answer.

DEMAND AND ITS LAW

Demand means the quantity of a good that buyers are willing and able to buy at each possible price in a given period. Both parts matter. A wish without money is not demand. A **demand schedule** is a table of prices and quantities. A **demand curve** is the same information drawn as a line.

Demand the quantity of a good that buyers are willing and able to buy at each price in a given period.

Demand needs both wish and ability to pay.

DAILY DEMAND FOR LOAVES IN THE NEIGHBOURHOOD

Price per loaf (Tk)	20	25	30	35	40
Quantity demanded	180	150	120	90	60

The table shows the **law of demand**. It states that, when the price of a good rises, the quantity demanded falls, and when the price falls, the quantity rises, if other things stay the same. Three reasons lie behind it. As price rises, we buy cheaper substitutes (the substitution effect). A higher price also reduces our real buying power (the income effect). And by the law of diminishing marginal utility, we will pay less for each extra unit.

Law of demand when the price of a good rises, the quantity demanded falls, other things equal.

Price up, quantity down.

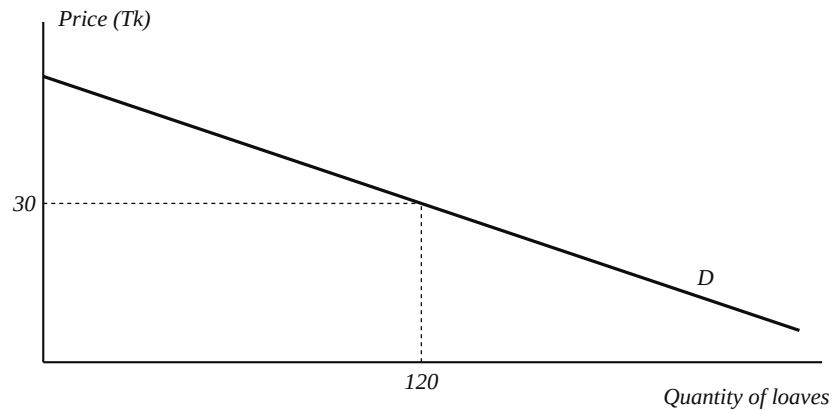


FIG. 3. The demand curve for loaves. It slopes downward from left to right.

Reading the figure.

To read the curve, pick a price on the vertical axis and go across to the line, then down to the quantity. At Tk 30 the quantity is 120 loaves. Moving along the curve, because of a change in price, is a **change in quantity demanded**. This is different from a shift of the whole curve.

FROM ONE BUYER TO THE MARKET

Adding up buyers.

A **market demand** curve is the sum of the demands of all buyers at each price. If two customers each buy 60 loaves at Tk 30, market demand at Tk 30 is 120 loaves. A price rise therefore reduces quantity both because each buyer buys less and because some buyers drop out. Throughout this book, “demand” means market demand unless we say otherwise.

Substitute goods goods that can replace each other, such as tea and coffee.

Complementary goods goods used together, such as bread and butter.

Related goods change demand.

Goods can be related. **Substitute goods** can replace each other, so a rise in the price of one raises demand for the other. **Complementary goods** are used together, so a rise in the price of one lowers demand for the other. Suppose the price of petrol rises. People drive less, and demand for cars, which are used with petrol, probably falls. Demand for bicycles, a substitute, probably rises.

SHIFTS OF THE CURVE

Other things that move demand.

The demand curve is drawn for given values of other things: the buyer’s income, tastes, the prices of related goods, expectations and the number of buyers. When one of these changes, the whole curve shifts. This is a **change in demand**. A rise in income usually moves the curve to the right: buyers want more at every price. A fall moves it to the left. Remember the difference: price moves us along the curve; every other cause shifts it.

EXCEPTIONS TO THE LAW

Giffen good a very basic good for which demand rises when its price rises.

Rare, but a real exception.

The law has a few exceptions. A **Giffen good** is a basic food for which quantity demanded rises when the price rises. Marshall (1895) credited Robert Giffen with the idea and linked it to the potato in nineteenth-century Ireland, though modern historians dispute the evidence. The reasoning is this: for a very poor family, a basic food takes most of its income. When its price rises, the family cannot afford better food, and it buys more of the basic food to stay fed.

Other exceptions.

Two other cases concern pride and crowds. The **snob effect** is the wish to own a good because few others have it: a rise in price may make a rare watch more desirable. The **bandwagon effect** is the wish to own what others own:

demand rises when many people are seen buying a popular phone. Expectations also matter. If buyers expect the price to rise further, they may buy more today. In each case the demand curve may slope upward over part of its range.

CASE FILE: DEMAND SCHEDULE, 1 MARCH 2024

Daily quantity demanded in the neighbourhood: Tk 20 gives 180 loaves, Tk 25 gives 150, Tk 30 gives 120, Tk 35 gives 90, Tk 40 gives 60. In equation form, $Q = 300 - 6P$. Check: $P = 30$ gives $Q = 300 - 180 = 120$.

SUMMARY. Demand is the quantity buyers are willing and able to buy at each price. The law says price and quantity move in opposite directions. A price change moves us along the curve; a change in income, taste, related prices or number of buyers shifts it. Giffen goods, snob and bandwagon effects are exceptions.

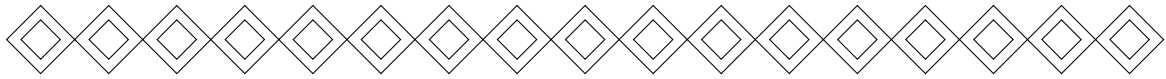
THINK IT THROUGH. From $Q = 300 - 6P$, find the quantity demanded at Tk 22. Then say whether a rise in the price of flour shifts this demand curve.

WHERE PEOPLE GO WRONG. “A fall in price shifts the curve to the right.” It does not. A price change moves us along the same curve; only other causes shift it.

IN HISTORY. Alfred Marshall’s *Principles* (1890) drew demand with price on the vertical axis and quantity on the horizontal. Economists have kept that convention for more than a century.

REVIEW QUESTIONS

1. State the law of demand.
2. Name two causes of a shift in demand.
3. What is a Giffen good?



CHAPTER THE FOURTH

ELASTICITY OF DEMAND

How much does demand react to a price change?

The chapter begins with two price rises that bring different results. It defines elasticity, measures price, income and cross elasticity, names the five forms, and lists the factors behind each.

Two shops each raise their price by 10 per cent. The bread shop loses almost no customers. The cake shop loses a third of them. Why do the same price rises give such different results? The law of demand tells us the direction of the change. It does not tell us the size. To measure size, we need a new tool.

Start with a question you can almost answer.

PRICE ELASTICITY OF DEMAND

Price elasticity of demand means the percentage change in quantity demanded divided by the percentage change in price. In symbols, $E_d = (\% \text{change in } Q) / (\% \text{change in } P)$. Percentage change means the change divided by the starting value, times 100. Because price and quantity move in opposite directions, the answer is negative. We usually drop the minus sign and speak of the size.

Price elasticity of demand the percentage change in quantity demanded divided by the percentage change in price.

Elasticity measures how strongly buyers react.

WORKED EXAMPLE. Price elasticity from a schedule

GIVEN	From the demand schedule of Chapter 3: at Tk 30 quantity is 120; at Tk 35 quantity is 90.
FIND	The price elasticity of demand for the rise from Tk 30 to Tk 35.
FORMUL A	$E_d = (\% \text{change in } Q) / (\% \text{change in } P)$.
STEPS	Change in $Q = 90 - 120 = -30$. $\% \text{change in } Q = -30/120 \times 100 = -25\%$. Change in $P = 35 - 30 = 5$. $\% \text{change in } P = 5/30 \times 100 = 16.7\%$. $E_d = -25 / 16.7 = -1.5$.
ANSWER	The size of elasticity is 1.5. Demand is elastic.
CHECK	Check: the percentage fall in quantity (25) is larger than the percentage rise in price (16.7), so the ratio must exceed 1. It does.

Now take a fall in price from Tk 25 to Tk 20. Quantity rises from 150 to 180. The change in Q is +30, which is 20 per cent of 150. The change in P is -5, which is 20 per cent of 25. So $E_d = 20 / (-20) = -1$. Take care: a common slip is to use the new value instead of the starting value, which gives a different answer. Always divide by the starting value, or use the midpoint if your teacher asks.

A second example, with messier numbers.

There are five forms. If E_d is greater than 1, demand is **elastic**: quantity changes by a larger percentage than price. If E_d is less than 1, demand is **inelastic**: quantity changes by a smaller percentage. If E_d equals 1, demand is **unitary elastic**. At the two extremes, demand is **perfectly inelastic** ($E_d = 0$,

Elastic demand quantity changes by a larger percentage than price.

Inelastic demand quantity changes by a smaller percentage than price.

Five forms, from zero to infinity.

quantity does not change at all) or **perfectly elastic** (E_d is infinite, buyers will buy any amount at one price and none above it).

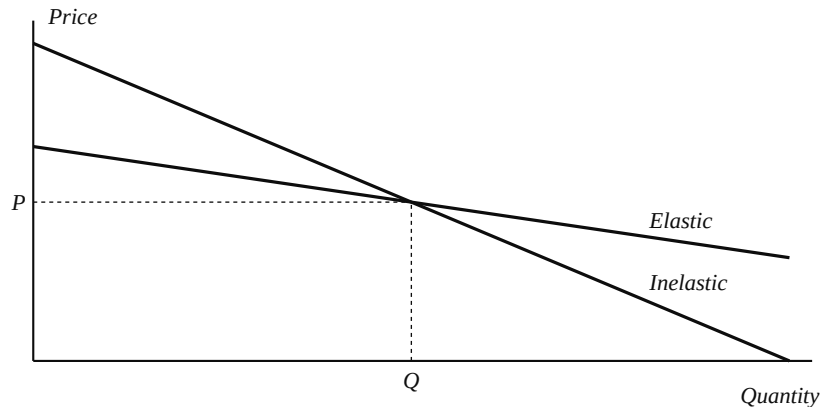


FIG. 4. Two demand curves through the same point. The flatter one is more elastic.

What decides elasticity?

Four factors matter. The more substitutes a good has, the more elastic its demand: if tea is dear, buyers switch to coffee. Necessities, such as salt, have inelastic demand; luxuries have elastic demand. A good that takes a large share of income is more elastic than one that takes a small share. And demand is more elastic over a long period, because buyers have time to find other goods.

INCOME AND CROSS ELASTICITY

Income elasticity of demand the percentage change in quantity demanded divided by the percentage change in income.

Income elasticity links buying to income.

Cross elasticity of demand the percentage change in quantity of one good divided by the percentage change in the price of another good.

Cross elasticity shows how goods are related.

Income elasticity of demand means the percentage change in quantity demanded divided by the percentage change in income. If income rises 10 per cent and cake purchases rise 20 per cent, the value is 2. A value above 1 marks a luxury; between 0 and 1 marks a necessity; below 0 marks an **inferior good**, one that people buy less of as they grow richer, such as very cheap coarse rice.

Cross elasticity of demand means the percentage change in the quantity of one good divided by the percentage change in the price of another. If a rival's bread price rises 10 per cent and our sales rise 5 per cent, the value is 0.5. A positive value shows that the goods are substitutes. A negative value shows that they are complements. A value near zero shows that the goods are unrelated.

WHY ELASTICITY MATTERS

Revenue and the price decision.

Total revenue means price multiplied by quantity sold. With elastic demand, a price cut raises revenue, because quantity rises by more than price falls. With inelastic demand, a price cut lowers revenue. A firm that knows its elasticity can choose the right price. So a seller of an inelastic good, such as salt, can raise its price with little loss of sales; a seller of an elastic good, such as cake, cannot.

CASE FILE: PRICE CHANGE TEST, 1 APRIL 2024

The bakery moves the loaf price from Tk 30 to Tk 35. Revenue before: $30 \times 120 =$ Tk 3,600. Revenue after: $35 \times 90 =$ Tk 3,150. Revenue falls, as the elasticity of 1.5 predicts. The bakery returns to Tk 30.

SUMMARY. Elasticity measures how much quantity responds to a change in price (or income, or another price). Demand is elastic above 1, inelastic below 1. Substitutes, necessity, share of income and time decide its size. With elastic demand, a price cut raises revenue.

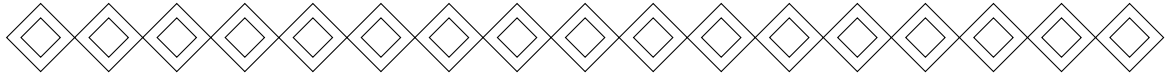
THINK IT THROUGH. A firm sells 200 units at Tk 10. It raises the price to Tk 11 and sells 150. Find the elasticity and say what happens to revenue.

WHERE PEOPLE GO WRONG. “A steep curve and an inelastic curve are the same.” Slope and elasticity are different, because elasticity uses percentages and slope uses units. A straight line has one slope but a different elasticity at each point.

IN HISTORY. Marshall introduced the measure of price elasticity of demand in 1890. He wanted a number that did not depend on units, so that the demand for wheat and the demand for cloth could be compared.

REVIEW QUESTIONS

1. Define price elasticity of demand.
2. $Q = 200$, $P = 10$; Q falls by 50 when P rises by 1. Find E_d .
3. What does a cross elasticity of -0.8 show?



CHAPTER THE FIFTH

CONSUMER SURPLUS AND INDIFFERENCE CURVE

Why does a bargain feel like a gift?

The chapter begins with a customer who pays less than she was willing to pay. It measures this gain as consumer surplus, then shows how indifference curves describe choice without counting utils.

In Chapter 2 a customer was ready to pay Tk 20 for her first bun. The price is only Tk 10. She walks away pleased. Where did the pleasure come from, and can we count it in taka? The answer gives economists a way to judge whether a market makes people better off.

Start with a question you can almost answer.

CONSUMER'S SURPLUS

Consumer's surplus means the difference between the most a buyer is willing to pay and the price actually paid. Dupuit (1844) first described the idea, and Marshall (1890) developed it. It rests on diminishing marginal utility: because each extra unit is worth less, only the last unit is worth exactly the price.

Consumer's surplus the difference between what a buyer is willing to pay and what the buyer actually pays.

The gain from paying less than you would.

WORKED EXAMPLE. Consumer's surplus from the survey

GIVEN	The customer values the buns at Tk 20, 16, 12, 8 and 4. The price is Tk 10.
FIND	Her consumer's surplus.
FORMULA	Surplus on a unit = value minus price. Buy a unit only if value is at least the price.
STEPS	She buys the first three buns, valued at 20, 16 and 12. Surplus: $(20 - 10) + (16 - 10) + (12 - 10) = 10 + 6 + 2 = 18$.
ANSWER	Her consumer's surplus is Tk 18.
CHECK	Check: she paid $3 \times 10 = \text{Tk } 30$ for buns she valued at $20 + 16 + 12 = \text{Tk } 48$. The gap is $48 - 30 = 18$. It agrees.

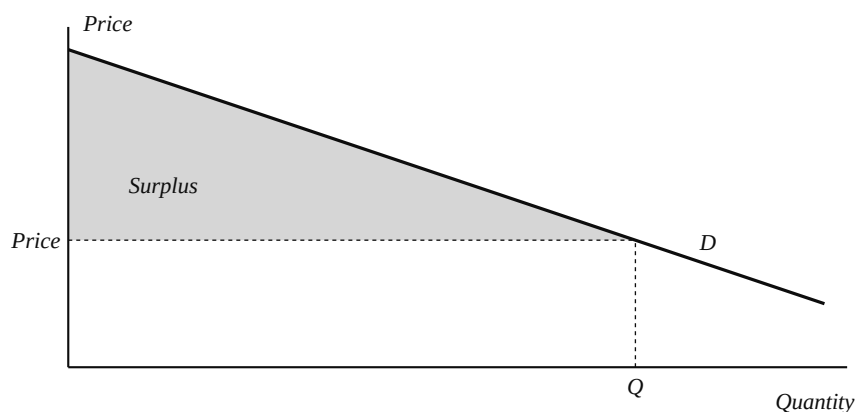


FIG. 5. Consumer's surplus is the shaded triangle between the demand curve and the price line.

Reading the figure.

The demand curve shows the most buyers would pay for each unit. The price line shows what they pay. The gap between the two, summed over all units bought, is the shaded area. If the price falls, the triangle grows, and buyers gain more. This is why a fall in price is a gain for buyers as a group.

Limits of the idea.

The measure has limits. It assumes that money has the same marginal utility for everybody, which is not exactly true. It cannot be applied to necessities, where the first unit is worth more than any sum. And it assumes there is no close substitute. Even so, it is a useful guide, and the next paragraph shows where.

Producer's surplus the difference between the price a seller receives and the lowest price the seller would accept.

The sellers' gain, as the mirror image.

Producer's surplus is the sellers' counterpart: the difference between the price received and the lowest price the seller would have accepted. Together, consumer's and producer's surplus measure the gain from trade. Governments use them to compare policies: a tax, a price ceiling or a monopoly reduces the sum, and the lost part is called the welfare loss. Public bodies use consumer's surplus to price water and power, and to weigh a new bridge against its cost.

CONSUMPTION AND CHOICE

Consumption the use of goods and services to satisfy wants.

Consumption ends the chain of production.

Consumption means the use of goods and services to satisfy wants. A consumer's budget is limited, so the consumer must choose among goods. Utility counting, with its utils, is hard to defend. The next method needs only ranking.

INDIFFERENCE CURVE ANALYSIS

Indifference curve a line showing all combinations of two goods that give a consumer equal satisfaction.

Every point on one curve is equally good.

Four properties.

An **indifference curve** means a line that joins all combinations of two goods that give the consumer the same satisfaction. If you are equally happy with 4 buns and 2 cups of tea, or 2 buns and 5 cups of tea, both points lie on one curve. The consumer is "indifferent" between them. Edgeworth (1881), Pareto (1906) and Hicks and Allen (1934) developed the tool.

Indifference curves have four properties. They slope downward, because more of one good must replace less of the other. They are convex to the origin, because of a falling **marginal rate of substitution** (MRS), the amount of one good given up for one more of the other. They never cross, since crossing would make one bundle both better and equal. And a curve higher and further from the origin shows greater satisfaction.

Budget line all combinations of two goods that the consumer's income can buy at given prices.

Income and prices draw the budget line.

A consumer cannot choose any point she likes. The **budget line** shows all combinations she can buy with her income at given prices. Suppose income is Tk 120, a bun costs Tk 10 and a cup of tea costs Tk 20. She can buy 12 buns and no

tea, or 6 cups and no buns, or any point between, such as 6 buns and 3 cups. In words: spending on buns plus spending on tea equals income.

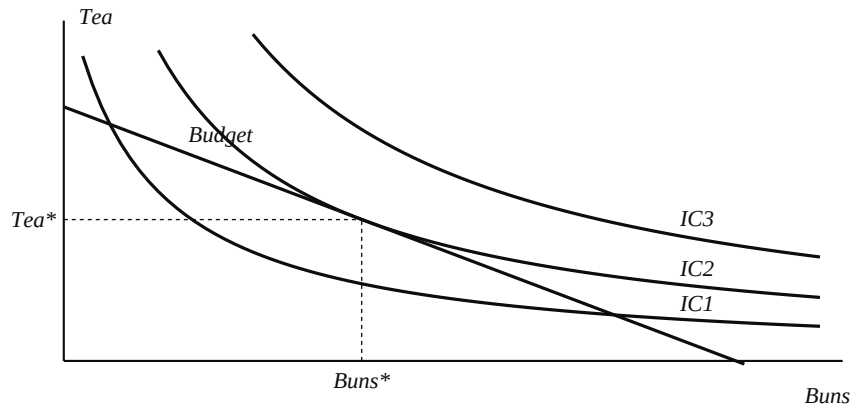


FIG. 6. The consumer chooses the point where the budget line just touches the highest curve she can reach.

The consumer reaches **equilibrium**, the point of best choice, where the budget line touches the highest indifference curve. At that point the two lines have the same slope. So the rate at which she is willing to swap goods (MRS) equals the rate at which the market lets her swap them (the price ratio). If the price of one good falls, the budget line turns outward, and she reaches a higher curve. The extra satisfaction can be valued in money, which links this method back to consumer's surplus.

Hicks's method measures the same surplus without utils. After a price fall, the consumer reaches a higher indifference curve. Now ask how much money we must take from her income to push her back to her old curve. That sum is her gain from the price fall, and so her consumer's surplus in money. The method needs only a ranking of bundles, not a count of satisfaction.

SUMMARY. Consumer's surplus is value minus price, summed over the units bought. Producer's surplus is its mirror. Indifference curves rank bundles without counting utils; the budget line shows what can be bought; the best choice is where the two touch.

THINK IT THROUGH. With income Tk 120, a bun at Tk 10 and tea at Tk 20, can the consumer buy 4 buns and 4 cups? Show your working.

WHERE PEOPLE GO WRONG. "Indifference curves can cross if the goods are similar." They cannot. If they crossed, one bundle would be equal to two others that are not equal to each other.

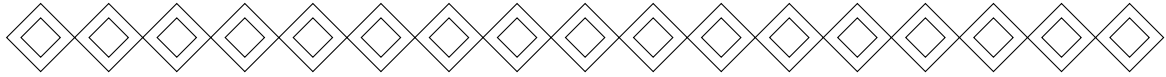
IN HISTORY. In 1844 the French engineer Jules Dupuit asked how to value a public bridge when it charged a toll. He saw that users would have paid more than the toll, and he measured the difference. This became consumer's surplus.

REVIEW QUESTIONS

1. Define consumer's surplus.
2. Give two properties of indifference curves.
3. With income Tk 100, bun Tk 10, how many buns if nothing else is bought?

Equilibrium of the consumer.

Measuring surplus with indifference curves.



CHAPTER THE SIXTH

THE THEORY OF SUPPLY

Why do sellers offer more at a higher price?

The chapter begins with a baker who bakes more when bread sells for more. It states the law of supply, lists what shifts supply, notes the exceptions, and measures the response by elasticity of supply.

In the spring of 2024 the price of bread in the neighbourhood rises. Rupali Bakery bakes more, and two other bakeries do the same. A higher price gives sellers a reason to offer more. This is the second half of every market story, and it mirrors the story of buyers.

Start with a question you can almost answer.

SUPPLY AND ITS LAW

Supply means the quantity of a good that sellers are willing and able to offer for sale at each price in a given period. A **supply schedule** is a table of prices and quantities. A **supply curve** is the same table drawn as a line. A schedule gives separate numbers; a curve joins them into one picture.

Supply the quantity of a good that sellers are willing and able to offer at each price in a given period.

Supply is the sellers' side.

DAILY SUPPLY OF LOAVES IN THE NEIGHBOURHOOD

Price per loaf (Tk)	20	25	30	35	40
Quantity supplied	60	90	120	150	180

The table shows the **law of supply**. It states that, when the price of a good rises, the quantity supplied rises, and when the price falls, the quantity falls, if other things stay the same. Two reasons lie behind it. A higher price makes it more profitable to produce. And producers face rising cost per unit when they expand, so they need a higher price to cover it. In equation form, $Q_s = 6P - 60$.

Law of supply when the price of a good rises, the quantity supplied rises, other things equal.

Price up, quantity up.

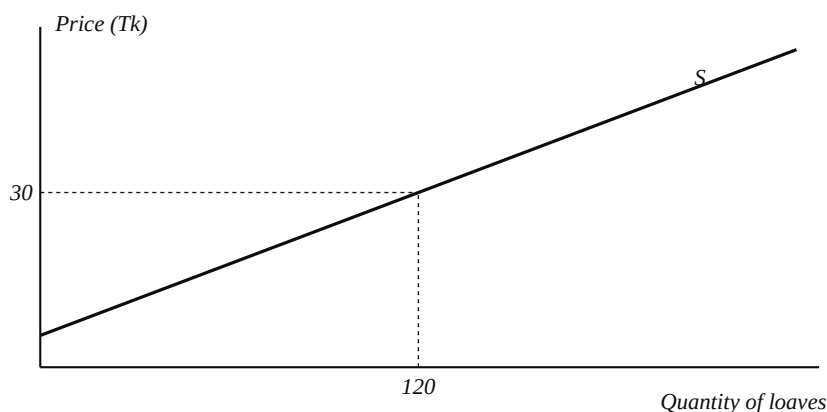


FIG. 7. The supply curve slopes upward from left to right.

WHAT MOVES THE SUPPLY CURVE

Determinants of supply.

A change in price moves us along the curve. Other causes shift the whole curve. The main ones are: the cost of inputs, such as flour and labour; technology, which lowers cost; taxes and subsidies; the prices of other goods the seller could make instead; the sellers' expectations of future prices; and the number of sellers. Cheaper flour moves the curve to the right: sellers offer more at every price. A tax on bread moves it to the left.

EXCEPTIONS TO THE LAW

Exceptional supply curve a supply curve that does not slope upward in the usual way.

A few cases break the rule.

An **exceptional supply curve** does not rise in the usual way. There are several cases. The supply of labour may bend backward: at very high wages, a worker may choose more leisure and work fewer hours. Goods with fixed supply, such as old paintings or land in a city, cannot rise with price. Perishable goods, such as fish at the end of the day, may be sold at any price rather than wasted. If sellers expect the price to rise further, they may hold stock back and offer less now.

ELASTICITY OF SUPPLY

Price elasticity of supply the percentage change in quantity supplied divided by the percentage change in price.

How strongly sellers respond.

Price elasticity of supply means the percentage change in quantity supplied divided by the percentage change in price. In symbols, $E_s = (\% \text{change in } Q_s) / (\% \text{change in } P)$. The answer is positive, because price and quantity move together. If E_s is above 1, supply is **elastic**; below 1, **inelastic**; at 1, unitary. If E_s is zero, supply is **perfectly inelastic**; if infinite, **perfectly elastic**.

WORKED EXAMPLE. Elasticity of supply

GIVEN	From the supply schedule: at Tk 30 supply is 120; at Tk 35 supply is 150.
FIND	E_s for the rise from Tk 30 to Tk 35.
FORMUL	$E_s = (\% \text{change in } Q_s) / (\% \text{change in } P)$.
A	
STEPS	Change in $Q_s = 150 - 120 = 30$. $\% \text{change} = 30/120 \times 100 = 25\%$. Change in $P = 5$. $\% \text{change} = 5/30 \times 100 = 16.7\%$. $E_s = 25 / 16.7 = 1.5$.
ANSWER	$E_s = 1.5$. Supply is elastic.
CHECK	Check: quantity rose by a larger percentage than price, so E_s is above 1. It is.

A neat fact: any straight supply line that starts at the point of origin has an elasticity of exactly 1 at every point. For such a line, quantity is a fixed multiple of price, so a 10 per cent rise in price gives a 10 per cent rise in quantity. A line that cuts the vertical axis (as our schedule does) has elasticity above 1. A line that cuts the horizontal axis has elasticity below 1.

A line from the origin.

Three factors matter. The first is spare capacity: a bakery with an idle oven can expand fast. The second is the ability to store output: goods that can be stored, such as rice, have elastic supply, and goods that spoil do not. The third is time: supply is inelastic in the very short run, when inputs are fixed, and more elastic in the long run, when firms can build and hire.

What decides the size?

CASE FILE: CAPACITY, 1 JUNE 2024

The bakery adds a second shift and can raise output from 120 to 150 loaves a day when the price rises from Tk 30 to Tk 35, as in the schedule. Its own elasticity of supply over that range is 1.5.

SUMMARY. Supply is the quantity sellers will offer at each price. The law says price and quantity rise together. Costs, technology, taxes, related prices, expectations and number of sellers shift the curve. Elasticity of supply measures the response; spare capacity, storability and time decide its size.

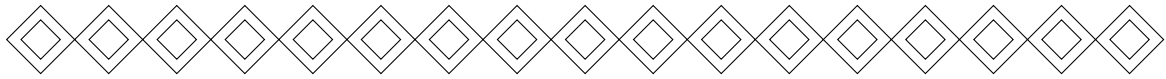
THINK IT THROUGH. Using $Q_s = 6P - 60$, find supply at Tk 25. Then say what happens to the curve if flour becomes cheaper.

WHERE PEOPLE GO WRONG. “A higher price shifts the supply curve to the right.” It does not; it moves us along the curve. Only a change in cost, technology and the like shifts it.

IN HISTORY. In 1890 Marshall compared supply and demand to the two blades of a pair of scissors. He said that it is as useless to ask which blade cuts as to ask whether price is set by supply or demand alone.

REVIEW QUESTIONS

1. State the law of supply.
2. Name two causes that shift supply to the right.
3. What is E_s of a straight supply line through the origin?



CHAPTER THE SEVENTH

MARKETS AND EQUILIBRIUM

How does a price find itself?

The chapter begins with a bakery that tries a price too high and a price too low. It defines a market, shows how supply and demand meet at an equilibrium price, describes the forces that bring it about, and asks whether lower prices are always good.

Suppose Rupali Bakery tries a loaf price of Tk 40 on Monday. At that price buyers want 60 loaves and sellers offer 180. On Tuesday it tries Tk 20. Buyers want 180 but sellers offer 60. Neither price lasts. On Wednesday the price settles at Tk 30, and buyers and sellers are both content. Why does Tk 30 work, and who set it?

Start with a question you can almost answer.

WHAT A MARKET IS

A **market** means any arrangement that brings buyers and sellers of a good together so that they can trade. It need not be a place. A phone call, a website or a bazaar can be a market. Markets differ by area (local, national, world), by time (very short, short and long period) and by the amount of competition. A **competitive market** has many buyers and many sellers, so that no one of them can set the price. Other kinds, monopoly and monopolistic competition, are studied in Chapter 10.

Market any arrangement that brings buyers and sellers of a good together to trade.

A market is a meeting, not necessarily a place.

EQUILIBRIUM

Equilibrium means a state in which nothing pushes the situation to change. In a market, the equilibrium price is the price at which quantity demanded equals quantity supplied. At that price, every buyer who wishes to buy finds a seller, and every seller finds a buyer.

Equilibrium a state in which quantity demanded equals quantity supplied, so there is no pressure for change.

Equilibrium is a point of rest.

WORKED EXAMPLE. Finding the equilibrium

GIVEN	Demand: $Q_d = 300 - 6P$. Supply: $Q_s = 6P - 60$.
FIND	The equilibrium price and quantity.
FORMULA	Set $Q_d = Q_s$.
STEPS	$300 - 6P = 6P - 60$. Add $6P$ to both sides and add 60 : $360 = 12P$. So $P = 360 / 12 = 30$. Put $P = 30$ in demand: $Q = 300 - 180 = 120$.
ANSWER	Equilibrium price is Tk 30 and quantity is 120 loaves.
CHECK	Check in supply: $Q_s = 6 \times 30 - 60 = 120$. Both sides agree.

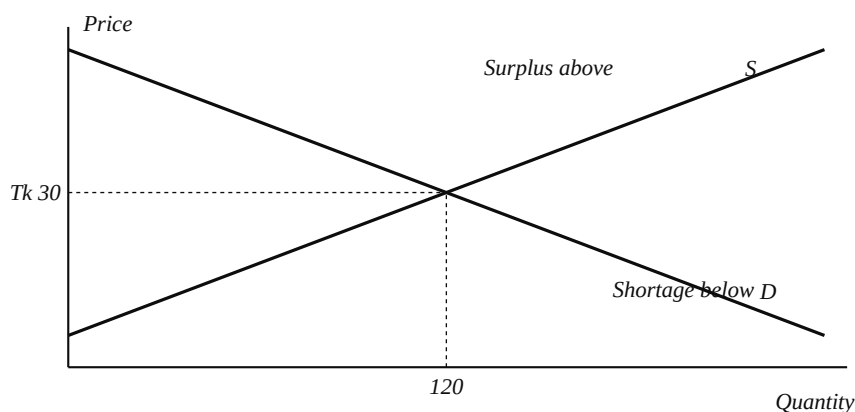


FIG. 8. The curves cross at the equilibrium. Above the price there is a surplus; below it, a shortage.

The forces at work.

Now see why other prices fail. At Tk 40, quantity supplied is 180 and quantity demanded is 60, so there is a **surplus** of 120 loaves. Sellers have unsold bread, and they cut the price. At Tk 20, quantity demanded is 180 and quantity supplied is 60, so there is a **shortage** of 120. Buyers compete, and the price rises. In both cases the price moves toward Tk 30. No one sets it. It emerges from many separate choices.

When a curve shifts.

Suppose a change in tastes adds 60 loaves to demand at every price, so $Q_d = 360 - 6P$. Then $360 - 6P = 6P - 60$, so $420 = 12P$ and $P = 35$. Quantity is $Q = 6 \times 35 - 60 = 150$. A rise in demand raises both price and quantity. A rise in supply does the opposite to price: it lowers price and raises quantity. Always find which curve moved, in which direction, and then read the new crossing.

PRICE IN DIFFERENT PERIODS

Time matters.

Marshall (1890) distinguished three periods. In the **market period**, supply is fixed, as with the morning's baked loaves, so demand alone sets the price. In the **short run**, firms can change output using their existing ovens, but cannot build new ones. In the **long run**, firms can enter and leave, and all inputs can change. Prices in the long run follow cost more closely, and prices in the short run follow demand more closely. Under other market types, as Chapter 10 shows, a firm may influence the price itself.

Is a lower price always good?

Not always. A lower price helps buyers, and consumer's surplus rises. But if the price falls below the cost of production, firms leave, supply shrinks and shortages follow. A government price ceiling set below equilibrium creates exactly the shortage described above. A very low price may also signal poor quality, or may be used by a large firm to drive out smaller rivals. A lower price is good for buyers only when it reflects lower cost and when the sellers remain.

CASE FILE: MARKET TEST, 4-6 MARCH 2024

Monday Tk 40: demand 60, supply 180, surplus 120. Tuesday Tk 20: demand 180, supply 60, shortage 120. Wednesday Tk 30: demand 120, supply 120. Revenue at the equilibrium: $30 \times 120 = \text{Tk } 3,600$ a day.

SUMMARY. A market brings buyers and sellers together. The equilibrium price makes quantity demanded equal to quantity supplied. A surplus pulls price down and a shortage pushes it up. A shift in demand or supply creates a new equilibrium. Lower prices are not always good.

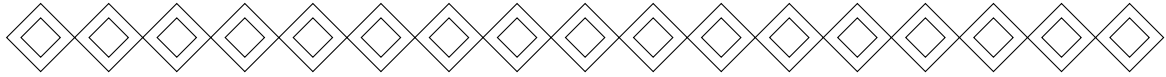
THINK IT THROUGH. Supply rises by 30 at every price, so $Q_s = 6P - 30$. Using $Q_d = 300 - 6P$, find the new price and quantity.

WHERE PEOPLE GO WRONG. “Equilibrium means everyone is happy.” It means only that plans match. Some buyers still cannot afford the good.

IN HISTORY. In the 1890 edition of his book, Marshall used diagrams where demand and supply cross, and called the crossing point the equilibrium. Before him, writers argued which one set the price.

REVIEW QUESTIONS

1. Define equilibrium price.
2. At a price above equilibrium, what exists and what happens to price?
3. $Q_d = 100 - 2P$, $Q_s = 3P - 20$. Find P .



CHAPTER THE EIGHTH

PRODUCTION AND RETURNS

What happens when we add one more baker?

The chapter begins with a bakery that hires workers one by one. It defines the production function, shows why extra workers add less and less, and distinguishes this from returns to scale.

The bakery has one oven. With one baker it makes 40 loaves a day. A second baker makes it possible to bake 100. A third raises output to 180. A seventh baker, however, makes output fall. Why should a helper ever reduce output? The answer explains the shape of every cost curve in the next chapter.

Start with a question you can almost answer.

THE PRODUCTION FUNCTION

Production means the making of goods and services that satisfy wants. Economists group the resources used into four **factors of production**: land (all natural resources), labour (human effort), capital (machines, buildings and tools) and enterprise (the skill of organising and taking risk).

The **production function** means the relationship between the inputs used and the greatest output they can make. In words: output depends on land, labour, capital and enterprise. In symbols, $Q = f(L, K)$, where Q is output, L is labour and K is capital. In the **short run**, at least one input is fixed. In the **long run**, all inputs can change.

Production the creation of goods and services that satisfy wants.

Production creates utility.

Production function the relation between the inputs used and the maximum output obtained.

Inputs in, output out.

THE LAW OF DIMINISHING RETURNS

Take the short run, with the oven fixed and labour variable. **Total product** (TP) is the total output. **Average product** (AP) is output per worker, TP divided by workers. **Marginal product** (MP) is the extra output from one more worker.

Three measures.

OUTPUT OF THE BAKERY WITH ONE OVEN

Bakers	Total product	Average product	Marginal product
1	40	40	40
2	100	50	60
3	180	60	80
4	240	60	60
5	280	56	40
6	300	50	20
7	290	41	-10

Law of diminishing returns when one input is added to fixed inputs, the extra output from each further unit eventually falls.

Eventually each extra worker adds less.

The table shows the **law of diminishing returns**. It states that, when more of one input is added to fixed amounts of the others, the marginal product of the added input eventually falls. Here it begins after the third baker, whose MP is 80. Turgot noted the effect in 1767, and Malthus, West and Ricardo stated it in 1815. Ricardo applied it to farmland.

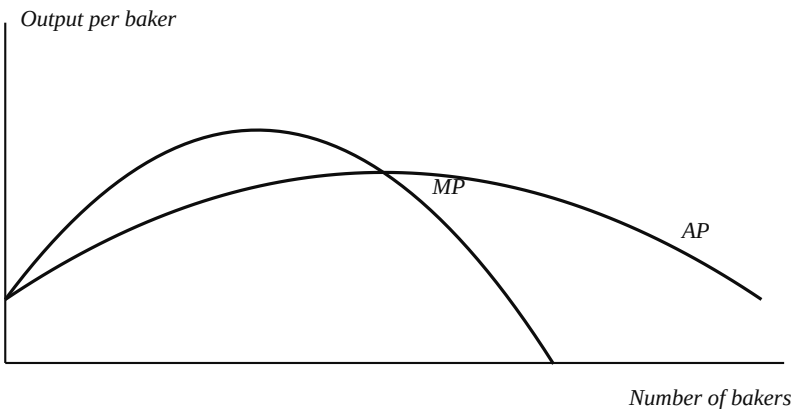


FIG. 9. Marginal product (MP) rises, then falls and cuts the average product (AP) curve at its highest point.

Why it happens.

The cause is simple. The oven is fixed. At first, extra bakers share tasks and use the oven better, so MP rises. Past a point, bakers crowd the kitchen and wait for the oven, so each adds less. The law holds if the extra units are of equal skill, if technology is unchanged and if at least one input is fixed. Output is in three stages: rising MP, falling but positive MP (where a sensible firm operates), and negative MP.

RETURNS TO SCALE

Returns to scale the change in output when all inputs are changed in the same proportion.

The long-run cousin of diminishing returns.

Returns to scale means the change in output when every input is changed in the same proportion. It is a long-run idea, since all inputs vary. Diminishing returns is a short-run idea, since one input is fixed. Do not confuse them. If output rises by a larger percentage than inputs, there are **increasing returns to scale**. If it rises by the same percentage, they are **constant**. If it rises by a smaller percentage, they are **decreasing**.

WORKED EXAMPLE. Types of returns to scale

GIVEN	Units of all inputs: 1, 2, 4, 8. Output: 100, 250, 500, 700.
FIND	Whether returns are increasing, constant or decreasing at each step.
FORMUL A	%change in output compared with %change in inputs.
STEPS	Step 1 to 2: inputs +100%; output 100 to 250 is +150%. Step 2 to 4: inputs +100%; output 250 to 500 is +100%. Step 4 to 8: inputs +100%; output 500 to 700 is +40%.
ANSWER	Increasing, then constant, then decreasing returns.
CHECK	Check: +150% exceeds +100%, +100% equals +100%, and +40% is below +100%. Each label fits.

Why returns to scale change.

At first, a larger firm gains economies: workers specialise, big machines work well, bulk buying lowers costs. Later, size brings strain: managers lose touch, orders move slowly and costs rise. Smith’s pin factory (1776) shows the first effect. Ford’s Model T line shows it too: the price fell from \$850 in 1908 to under \$300 by the mid-1920s, as assembly-line output grew.

CASE FILE: STAFFING, 1 SEPT 2024

The bakery sets four bakers, since MP is still 60 and AP is at its highest (60). A fifth baker would add only 40 loaves, which is less than the average. The owner leaves the number at four.

SUMMARY. The production function links inputs to output. In the short run, extra units of a variable input eventually add less. In the long run, doubling all inputs may more than double, exactly double or less than double output.

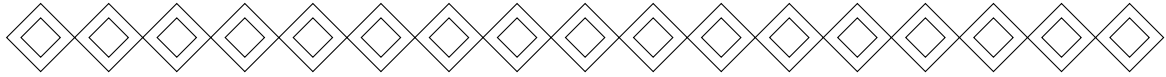
THINK IT THROUGH. Using the table, find MP of the fifth baker, and AP with six bakers. Why does the owner stop at four, and not seven?

WHERE PEOPLE GO WRONG. “Diminishing returns means total output falls.” Not at first. Total product still rises while MP is positive.

IN HISTORY. In 1767 the French minister Turgot saw that adding more labour to a fixed plot of land gave smaller and smaller extra harvests. Ricardo used the same idea to explain rent.

REVIEW QUESTIONS

1. Define the production function.
2. Distinguish diminishing returns from returns to scale.
3. Inputs double and output rises 60%. What returns?



CHAPTER THE NINTH

THE COSTS OF PRODUCTION

What does a loaf really cost?

The chapter begins with a baker who thinks her own time is free. It separates explicit and implicit costs, builds the cost curves from a table, compares short and long run, and finds the cheapest mix of inputs.

The owner of Rupali Bakery adds up her bills for flour, rent, wages and gas, and calls the total her cost. She has left out her own work and the money she put into the shop. Is her real cost larger than the bills? Economists say yes, and the difference changes the answer to the question “Am I making a profit?”

Start with a question you can almost answer.

OPPORTUNITY COST AND KINDS OF COST

Recall from Chapter 1 that **opportunity cost** is the value of the best alternative given up. To measure it, ask what the same resources would earn in their next best use. In the bakery, one cake needs the oven for as long as five loaves. So the opportunity cost of one cake is 5 loaves, and of one loaf it is 1/5 of a cake, that is, 0.2 cake. The cost is stated in goods, not only in money.

Opportunity cost, measured.

An **explicit cost** is a payment to an outsider, such as for flour or rent. An **implicit cost** is the value of the owner’s own resources that are used in the business. If the owner could earn Tk 1,000 a day in another job, that sum is an implicit cost of her time. If her savings could earn Tk 500 a day elsewhere, that is an implicit cost of her capital. Two examples of implicit cost are therefore the owner’s time and the owner’s own money.

Explicit cost a cost that needs a payment to someone outside the firm.

Implicit cost the value of resources the owner supplies, for which no payment is made.

Two kinds of cost: paid and unpaid.

Costs also differ by how they react to output. A **fixed cost** (FC) does not change with output in the short run, such as rent and the oven. A **variable cost** (VC) changes with output, such as flour and gas. **Total cost** (TC) is fixed plus variable cost. Each cost can also be stated per unit: **average cost** (AC) is TC divided by output, and **marginal cost** (MC) is the extra cost of one more unit.

Fixed cost a cost that does not change with output in the short run.

Variable cost a cost that changes as output changes.

Fixed does not move; variable does.

COSTS OF THE BAKERY, IN TAKA PER DAY (FC = 1,500)

Loaves	VC	TC	AC	AVC	MC
100	2,400	3,900	39.0	24.0	24
200	4,200	5,700	28.5	21.0	18
300	5,700	7,200	24.0	19.0	15
400	7,600	9,100	22.8	19.0	19
500	10,500	12,000	24.0	21.0	29

WORKED EXAMPLE. Cost per loaf

GIVEN	At 300 loaves, FC = Tk 1,500 and VC = Tk 5,700. Going from 300 to 400 loaves raises VC to Tk 7,600.
FIND	AC at 300 loaves, and MC of the step from 300 to 400.
FORMUL	$AC = TC / Q$. $MC = \text{change in } TC / \text{change in } Q$.
A	
STEPS	$TC = 1,500 + 5,700 = 7,200$. $AC = 7,200 / 300 = 24$. At 400: $TC = 1,500 + 7,600 = 9,100$. Change in $TC = 9,100 - 7,200 = 1,900$. $MC = 1,900 / 100 = 19$.
ANSWER	$AC = \text{Tk } 24$ a loaf; $MC = \text{Tk } 19$ a loaf.
CHECK	Check: MC (19) is below AC (24), so AC should be falling as output rises to 400. In the table it falls to 22.8.

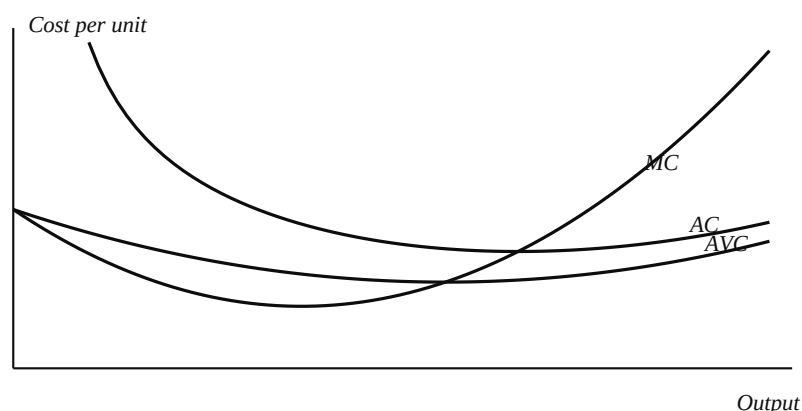


FIG. 10. Short-run cost curves. MC cuts both AVC and AC at their lowest points.

The pattern.

Look at the curves. AC and AVC fall at first and then rise, so each is U-shaped. AC falls at first because fixed cost is spread over more loaves. It rises later because of diminishing returns (Chapter 8). MC rises faster than the others and cuts them at their lowest points. A good rule: if MC is below the average, the average falls; if MC is above, the average rises. A marginal figure pulls the average in its direction.

SHORT RUN AND LONG RUN

Long-run average cost curve the lowest cost of each output when all inputs can change.

In the long run, no cost is fixed.

In the short run, the bakery has one oven, so it chooses a cost curve for that size. In the long run, it can choose a larger or smaller oven. The **long-run average cost curve** (LAC) joins the lowest points available from all the short-run curves, so it touches each of them from below. It is U-shaped as well: it falls with economies of scale and rises with diseconomies of scale (Chapter 8).

LEAST-COST COMBINATION

Least-cost combination the mix of inputs that produces a given output at the lowest total cost.

Spend each taka where it buys most.

A firm can often make the same output with different mixes of labour and machines. The **least-cost combination** is the mix that gives a target output at the lowest cost. The rule is: the extra output per taka should be equal for every input. In words, marginal product of labour divided by the wage must equal marginal product of capital divided by its hire price.

WORKED EXAMPLE. Least-cost mix

GIVEN	A baker's wage is Tk 800 a day; a machine hire is Tk 400 a day. Extra output: baker 40 loaves; machine 24 loaves.
FIND	Which input is the better buy.
FORMUL A	MP of labour / wage compared with MP of capital / hire price.
STEPS	Baker: $40 / 800 = 0.05$ loaf per taka. Machine: $24 / 400 = 0.06$ loaf per taka.
ANSWER	Machines give more per taka, so hire more machines and fewer bakers.
CHECK	Check: as machines increase, their MP falls (diminishing returns), until the two ratios are equal. Then no further change pays.

CASE FILE: COSTS, 1 OCT 2024

At 500 loaves a day: total cost Tk 12,000 (explicit). Implicit costs: owner's time Tk 1,000, own capital Tk 500. Full economic cost: Tk 13,500.

SUMMARY. Opportunity cost is the best alternative given up. Costs may be explicit or implicit, fixed or variable. AC and MC are U-shaped; MC cuts AC at its minimum. The long-run curve allows all inputs to change. The cheapest mix equalises extra output per taka.

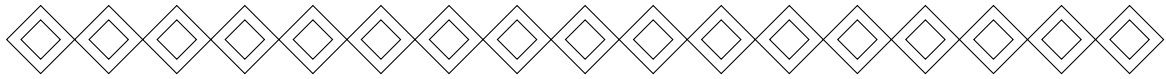
THINK IT THROUGH. If FC is Tk 1,500 and VC at 200 loaves is Tk 4,200, find AFC, AVC and AC. Which cost is the owner's own time?

WHERE PEOPLE GO WRONG. "Sunk and fixed costs should change the price." In the short run, fixed costs do not change with output, so they do not affect the choice of how much to bake.

IN HISTORY. The term "opportunity cost" was first used by Friedrich von Wieser in 1914. Before him, Ricardo had already compared the cost of a good with the other goods that the same labour could make.

REVIEW QUESTIONS

1. What is opportunity cost?
2. Name two implicit costs.
3. $AC = 24$, $MC = 29$. Does AC rise or fall?



CHAPTER THE TENTH

MARKET STRUCTURES AND THE FIRM

Who decides the price: the seller or the market?

The chapter begins with a baker who must accept the price and a monopolist who may choose it. It defines market structure, derives equilibrium for the competitive firm and industry, and compares monopoly and monopolistic competition.

The bakery cannot sell a loaf at Tk 50 when the shop next door sells at Tk 30, because buyers walk away. The only water company in a small town can raise its price and still sell. What separates the two sellers? Not their wish for profit, but the number of rivals they face and how alike their goods are.

Start with a question you can almost answer.

MARKET STRUCTURE

Market structure means the features of a market that decide how firms behave: the number of sellers, whether the goods are alike, and how easily new firms can enter. Four main types exist. Price-setting power rises as we move down the table.

Market structure the features of a market that decide how firms behave, mainly the number of sellers, the kind of product and the ease of entry.

Competition decides who sets the price.

FOUR MARKET STRUCTURES

Type	Sellers	Product	Entry	Control of price
Perfect competition	Very many	Identical	Free	None
Monopolistic competition	Many	Differentiated	Free	Some
Oligopoly	Few	Alike or differentiated	Hard	Considerable
Monopoly	One	No close substitute	Blocked	Large

PERFECT COMPETITION

Perfect competition means a market with very many small buyers and sellers, identical goods, free entry and exit, and full knowledge of prices. **Pure competition** is a narrower idea: it needs only the first three, with no element of monopoly, and it does not need full knowledge or free movement of resources. In perfect competition, each firm is a **price taker**. It accepts the market price, so its extra revenue from one more unit (marginal revenue, MR) equals the price.

Perfect competition a market with very many small buyers and sellers, identical goods, free entry and full information.

Price is set by the market.

Equilibrium of the firm.

A firm that wishes to earn the most profit expands output as long as the extra unit adds more to revenue than to cost. It stops where MR equals MC, with MC rising. For the competitive firm, MR equals price, so the rule is $P = MC$. In the short run the firm may earn a profit, break even or make a loss. It continues to produce in the short run if price covers at least average variable cost.

WORKED EXAMPLE. Short-run profit of the firm

GIVEN	Market price Tk 30. Costs from Chapter 9: at 500 loaves, $TC = \text{Tk } 12,000$ and $MC = \text{Tk } 29$.
FIND	The output that maximises profit, and the profit.
FORMUL A	Rule: $P = MC$. Profit = $(P - AC) \times Q$.
STEPS	At 400 loaves MC is Tk 19, below the price; the firm gains by baking more. At 500 loaves MC is Tk 29, just below Tk 30, so 500 is the best output on the table. $AC = 12,000 / 500 = 24$. Profit = $(30 - 24) \times 500 = 6 \times 500 = 3,000$.
ANSWER	The firm bakes 500 loaves and earns Tk 3,000 a day.
CHECK	Check: revenue $30 \times 500 = 15,000$; cost 12,000; the difference is 3,000. It matches.

The industry in the short and long run.

The **industry** means all firms that make the same good. In the short run, the industry is in equilibrium when market demand equals the sum of the supplies of all firms, which fixes the price each firm accepts. In the long run, profit attracts new firms. Their entry raises supply and lowers price. Entry continues until price equals the lowest AC, and each firm earns only **normal profit**, the minimum reward needed to keep it in business. A loss has the opposite effect: firms leave, supply falls, and price rises. The industry reaches long-run equilibrium when no firm wishes to enter or leave.

MONOPOLY

Monopoly a market with a single seller of a good that has no close substitute.

One seller sets the price.

Monopoly means a market with only one seller and no close substitute. A **pure monopoly** is the extreme case. A **bilateral monopoly** has one seller and also only one buyer, such as a single union facing a single employer. The monopolist faces the whole market demand, so to sell more it must lower the price on all units. Its marginal revenue is therefore below price. It maximises profit where MR equals MC, then charges the highest price buyers will pay for that output, which is above MC.

MONOPOLISTIC COMPETITION

Monopolistic competition a market with many sellers of similar but not identical goods, and free entry.

Many sellers, each with a small monopoly.

Monopolistic competition means a market with many sellers whose goods are similar but not identical, with free entry. Chamberlin and Robinson described it in 1933. **Product differentiation** is its typical feature. It means that each firm makes a version of the good that buyers see as different, through brand, quality, taste or location. Rupali’s loaf, with its own recipe and shop, is one example. Differentiation gives each firm a small monopoly and a downward-sloping demand curve.

Short and long run.

In the short run, the firm acts like a monopolist: it sets MR equal to MC and may earn a profit. In the long run, profit invites new firms with close substitutes. The firm’s demand shifts left and becomes flatter, until the demand curve just touches the AC curve. Price equals AC, and the firm earns only normal profit. Yet the output is below the point of lowest cost. This is called **excess capacity**.

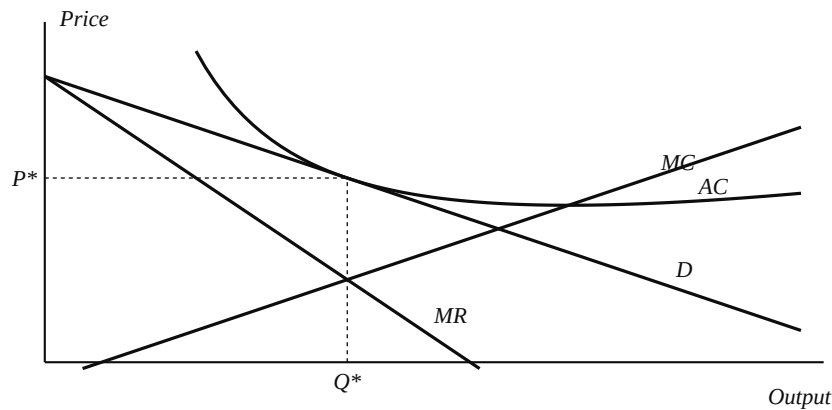


FIG. 11. Long-run equilibrium under monopolistic competition. The demand curve (D) touches AC at one point, and MR meets MC at the same output.

MARKET STRUCTURE AND PRICING

Structure decides whether a firm takes the price or makes it. A perfect competitor takes it. A monopolist and a monopolistic competitor choose it, within the limit of demand. An **oligopolist**, one of a few sellers, must also guess how rivals will answer a price change, so prices tend to stay stable. Many firms aim for goals other than profit, such as sales, market share or growth. Profit is still the most useful single guide, but it is not the only goal.

The role of structure in pricing.

CASE FILE: MARKET POSITION, 1 JAN 2025

The bakery sells a loaf with its own recipe, and the shop two streets away sells a similar one. Rupali Bakery therefore faces monopolistic competition. It has some control over price, and any profit above normal draws in new bakeries.

SUMMARY. Market structure depends on number of sellers, kind of product and entry. A competitive firm sets $P = MC$; entry drives profit to normal in the long run. A monopolist sets $MR = MC$ and charges a price above MC . Monopolistic competition has differentiation and, in the long run, normal profit and excess capacity.

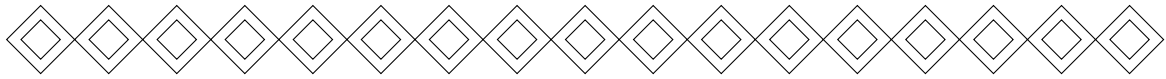
THINK IT THROUGH. At Tk 30, would the firm of Chapter 9 bake 300 loaves? Compare price with MC at 300 and say whether more or less output would raise profit.

WHERE PEOPLE GO WRONG. “Perfect and pure competition are the same.” They are not. Perfect competition adds full knowledge and perfect mobility to the conditions of pure competition.

IN HISTORY. In 1933, two books appeared in the same year, by Edward Chamberlin in the United States and Joan Robinson in England. Each showed that most real firms are neither perfect competitors nor pure monopolists.

REVIEW QUESTIONS

1. Define market structure.
2. State the profit-maximising rule of a competitive firm.
3. What is product differentiation?



CHAPTER THE ELEVENTH

RENT, WAGES, INTEREST AND PROFIT

Who gets what the bakery earns?

The chapter begins with the bakery's daily takings and asks who receives them. It explains how rent, wages, interest and profit are fixed, using Ricardo's theory, the marginal productivity idea and the main theories of profit.

On a good day Rupali Bakery takes Tk 15,000. Part goes to the landlord, part to the bakers, part to the lender who financed the oven, and part is left for the owner. What decides each share? Economists call this question **distribution**: the sharing of the product among the owners of land, labour, capital and enterprise.

Start with a question you can almost answer.

RENT

Economic rent means the payment to a factor whose supply is fixed, in excess of what is needed to keep it in use. Land is the main example, since the nation's land cannot be increased by a higher price. Ricardo (1817) built a theory on it. Rent is different from the everyday word, which means a payment to a landlord for a house or shop.

Economic rent the payment to a factor in fixed supply, above what is needed to keep it in use.

Rent is the reward for what is in fixed supply.

WORKED EXAMPLE. Ricardo's theory

GIVEN	Three plots of equal size, each worked with the same cost. Wheat yields: plot A 40, plot B 30, plot C 20 units.
FIND	The rent of each plot.
FORMUL A	Rent = yield of the plot minus yield of the marginal (worst) plot used.
STEPS	Plot C is the marginal plot: it only covers its costs, so its rent is 0. Rent of B = $30 - 20 = 10$. Rent of A = $40 - 20 = 20$.
ANSWER	Rents: A 20, B 10, C 0 units of wheat.
CHECK	Check: rent measures the extra advantage over the worst land. A has the largest advantage (20), so it has the largest rent.

Ricardo said that rent is a **differential surplus**: it arises from differences in fertility and location. On the marginal land there is no rent, so rent does not enter the price of grain. Critics reply that land has alternative uses, so for the economy as a whole it does have a cost. The modern view agrees with Ricardo for a single use, and adds that rent can be viewed as a cost for a user who must outbid others. **Quasi-rent** is a temporary surplus on a man-made asset, such as a machine, until supply can be increased. **Artificial rent** is a surplus created by restrictions, such as a licence, and it differs from the real rent of land.

Differential surplus the extra produce of better land over the worst land in use.

Rent is a surplus, not a cost on the margin.

WAGES

Nominal wage the wage paid in money.

Real wage what the money wage can buy.

Real wages count what buys.

The **nominal wage** is the wage in money. The **real wage** is what that money can buy, found by correcting for prices. The formula is: real wage = nominal wage divided by the price index, times 100.

WORKED EXAMPLE. Nominal and real wage

GIVEN	A baker's wage rises from Tk 800 to Tk 880. The price index rises from 100 to 120.
FIND	The new real wage, and whether the worker gained.
FORMUL A	Real wage = nominal wage / price index x 100.
STEPS	Real wage = $880 / 120 \times 100 = 733.3$ (in base-year taka). Before: $800 / 100 \times 100 = 800$.
ANSWER	The real wage fell from Tk 800 to Tk 733.
CHECK	Check: wages rose 10% but prices rose 20%, so buying power must have fallen. It did.

Why wages vary.

Wages differ for reasons that are simple to state. Skilled work needs training, which costs time and money. Dangerous or unpleasant work needs extra pay. Some jobs are steady and others are not. Trade unions and law may raise some wages. Wages differ between regions because of different costs of living and different supply of labour. A job that gives status may pay less, because people want it for other reasons.

How wages are determined.

In a competitive labour market, wages are set by demand and supply. The employer's demand rests on **marginal productivity**: a worker is hired if the value of the extra output is at least the wage. In the bakery, each loaf adds Tk 15 of value after materials. The fourth baker adds $60 \times 15 = \text{Tk } 900$, above the wage of 800, so is hired. The fifth adds $40 \times 15 = \text{Tk } 600$, below 800, so is not. Labour supply and bargaining also matter, so the result may differ.

INTEREST

Interest the payment for the use of borrowed money over a period of time.

Interest is the price of waiting.

Interest means the payment for the use of borrowed money for a period. Lenders give up spending now and take risk, and interest rewards them. In the **loanable funds** view, the interest rate is set where the supply of savings meets the demand for loans. Keynes (1936) added that people hold money for safety and convenience, so interest is the price of giving up that liquidity. A loan of Tk 100,000 at 10 per cent costs Tk 10,000 a year. If prices rise 6 per cent, the **real interest rate** is about $10 - 6 = 4$ per cent.

PROFIT

Economic profit total revenue minus all costs, explicit and implicit.

Accounting profit total revenue minus explicit costs only.

Two ways to count profit.

Accounting profit is revenue minus explicit costs. **Economic profit** is revenue minus all costs, including implicit ones. Using the bakery's figures at 500 loaves: revenue Tk 15,000 less explicit costs Tk 12,000 gives an accounting profit of Tk 3,000. Subtract the implicit costs of Tk 1,500 and the economic profit is Tk 1,500. **Normal profit** is the reward needed to keep the owner in business; it is already counted among the implicit costs.

Four main theories answer. Knight (1921) said profit rewards the bearing of **risk**, the uncertainty that cannot be insured. The marginal productivity theory treats the entrepreneur as one more factor, paid for the extra output she brings. Schumpeter (1911) said profit rewards **innovation**, the new method or product, and falls away as rivals copy it. A fourth view is that monopoly power or luck may also give surplus profit, which does not reward effort.

Why profit occurs.

CASE FILE: THE DAY'S TAKINGS, 1 OCT 2024

Takings Tk 15,000: materials, gas and other running costs; four bakers at Tk 800 = 3,200; rent; interest on the oven loan; and the rest to the owner. The owner's economic profit is Tk 1,500, as computed above.

SUMMARY. Rent rewards a factor in fixed supply and is a differential surplus on land. Wages depend on marginal productivity, supply and bargaining; real wages correct for prices. Interest rewards waiting and is set by loanable funds and liquidity. Profit rewards risk, innovation and enterprise, and economic profit is smaller than accounting profit.

THINK IT THROUGH. Plot A yields 50, B 35 and C 25, all at the same cost. Find the rent of A and B by Ricardo's method.

WHERE PEOPLE GO WRONG. "A higher nominal wage means a higher real wage." Not if prices rise faster. Always compare the two percentage changes.

IN HISTORY. In 1815 Ricardo wrote about the Corn Laws, the duties on imported grain in Britain, and argued that they raised rents of landowners at the cost of the rest. He used the theory of rent to make the case.

REVIEW QUESTIONS

1. Define economic rent.
2. Distinguish nominal wage from real wage.
3. Revenue 10,000, explicit costs 7,000, implicit costs 1,200. Find both profits.

ANSWERS TO THE REVIEW QUESTIONS

CHAPTER I

1. Because every person and every country has limited resources compared with wants, whether rich or poor.
2. Micro: why does a loaf cost Tk 30? Macro: why are prices across the nation rising?
3. Normative. It says what ought to be and rests on values; facts alone cannot settle it.

CHAPTER II

1. The extra satisfaction a person gets from one more unit of a good.
2. 60 minus 50 = 10 utils.
3. Units of the same kind and reasonable size; no change in taste, income or prices.

CHAPTER III

1. When the price of a good rises, quantity demanded falls, other things equal.
2. Any two of: income, tastes, prices of related goods, expectations, number of buyers.
3. A basic good for which demand rises when its price rises.

CHAPTER IV

1. The percentage change in quantity demanded divided by the percentage change in price.
2. %change Q = -25%; %change P = 10%; Ed = 2.5, elastic.
3. The two goods are complements.

CHAPTER V

1. The difference between what a buyer would pay and what is paid.
2. They slope downward and are convex to the origin; they never cross.
3. 10 buns.

CHAPTER VI

1. When price rises, quantity supplied rises, other things equal.
2. Cheaper inputs, better technology, a subsidy, more sellers.
3. Exactly 1 at every point.

CHAPTER VII

1. The price at which quantity demanded equals quantity supplied.
2. A surplus; price falls.
3. $100 - 2P = 3P - 20$, so $120 = 5P$ and $P = 24$.

CHAPTER VIII

1. The relationship between inputs and the greatest output they can produce.
2. Diminishing returns: one input varies, others fixed (short run). Returns to scale: all inputs vary together (long run).
3. Decreasing returns to scale.

CHAPTER IX

1. The value of the best alternative given up.
2. The owner's time and the owner's own capital.
3. Rise, because MC is above AC.

CHAPTER X

1. The features of a market, such as number of sellers, product type and entry, that decide how firms behave.
2. Produce where price equals marginal cost, with MC rising.
3. The making of goods that are similar but seen by buyers as different.

CHAPTER XI

1. A payment to a factor in fixed supply, above what is needed to keep it in use.
2. Nominal wage is in money; real wage is what that money can buy.
3. Accounting profit 3,000; economic profit 1,800.

GLOSSARY AND INDEX

Each entry gives the term, its meaning, and the chapter and page of first use.

Accounting profit: total revenue minus explicit costs only. *Ch. 11, p. 46.*

Budget line: all combinations of two goods that the consumer's income can buy at given prices. *Ch. 5, p. 22.*

Complementary goods: goods used together, such as bread and butter. *Ch. 3, p. 14.*

Consumer's surplus: the difference between what a buyer is willing to pay and what the buyer actually pays. *Ch. 5, p. 21.*

Consumption: the use of goods and services to satisfy wants. *Ch. 5, p. 22.*

Cross elasticity of demand: the percentage change in quantity of one good divided by the percentage change in the price of another good. *Ch. 4, p. 18.*

Demand: the quantity of a good that buyers are willing and able to buy at each price in a given period. *Ch. 3, p. 13.*

Differential surplus: the extra produce of better land over the worst land in use. *Ch. 11, p. 45.*

Economic law: a statement of the usual relation between economic events, true when other things stay equal. *Ch. 1, p. vi.*

Economic profit: total revenue minus all costs, explicit and implicit. *Ch. 11, p. 46.*

Economic rent: the payment to a factor in fixed supply, above what is needed to keep it in use. *Ch. 11, p. 45.*

Economic system: the way a society organises the choices of what, how and for whom to produce. *Ch. 1, p. v.*

Economics: the study of how people choose to use limited resources to satisfy wants. *Ch. 1, p. iv.*

Elastic demand: quantity changes by a larger percentage than price. *Ch. 4, p. 17.*

Equilibrium: a state in which quantity demanded equals quantity supplied, so there is no pressure for change. *Ch. 7, p. 29.*

Exceptional supply curve: a supply curve that does not slope upward in the usual way. *Ch. 6, p. 26.*

Explicit cost: a cost that needs a payment to someone outside the firm. *Ch. 9, p. 37.*

Fixed cost: a cost that does not change with output in the short run. *Ch. 9, p. 37.*

Giffen good: a very basic good for which demand rises when its price rises. *Ch. 3, p. 14.*

Implicit cost: the value of resources the owner supplies, for which no payment is made. *Ch. 9, p. 37.*

Income elasticity of demand: the percentage change in quantity demanded divided by the percentage change in income. *Ch. 4, p. 18.*

Indifference curve: a line showing all combinations of two goods that give a consumer equal satisfaction. *Ch. 5, p. 22.*

Inelastic demand: quantity changes by a smaller percentage than price. *Ch. 4, p. 17.*

Interest: the payment for the use of borrowed money over a period of time. *Ch. 11, p. 46.*

Law of demand: when the price of a good rises, the quantity demanded falls, other things equal. *Ch. 3, p. 13.*

Law of diminishing marginal utility: as a person consumes more units, the extra satisfaction from each further unit falls. *Ch. 2, p. ix.*

Law of diminishing returns: when one input is added to fixed inputs, the extra output from each further unit eventually falls. *Ch. 8, p. 34.*

Law of supply: when the price of a good rises, the quantity supplied rises, other things equal. *Ch. 6, p. 25.*

Least-cost combination: the mix of inputs that produces a given output at the lowest total cost. *Ch. 9, p. 38.*

Long-run average cost curve: the lowest cost of each output when all inputs can change. *Ch. 9, p. 38.*

Macroeconomics: the study of the whole economy, such as total output, prices and jobs. *Ch. 1, p. v.*

Marginal utility: the extra satisfaction from one more unit. *Ch. 2, p. viii.*

Market: any arrangement that brings buyers and sellers of a good together to trade. *Ch. 7, p. 29.*

- Market structure:** the features of a market that decide how firms behave, mainly the number of sellers, the kind of product and the ease of entry. *Ch. 10, p. 41.*
- Microeconomics:** the study of the choices of single households, firms and markets. *Ch. 1, p. v.*
- Monopolistic competition:** a market with many sellers of similar but not identical goods, and free entry. *Ch. 10, p. 42.*
- Monopoly:** a market with a single seller of a good that has no close substitute. *Ch. 10, p. 42.*
- Nominal wage:** the wage paid in money. *Ch. 11, p. 46.*
- Opportunity cost:** the value of the best alternative given up when a choice is made. *Ch. 1, p. v.*
- Perfect competition:** a market with very many small buyers and sellers, identical goods, free entry and full information. *Ch. 10, p. 41.*
- Positive statement:** a statement about what is, which facts can test. *Ch. 1, p. vi.*
- Price elasticity of demand:** the percentage change in quantity demanded divided by the percentage change in price. *Ch. 4, p. 17.*
- Price elasticity of supply:** the percentage change in quantity supplied divided by the percentage change in price. *Ch. 6, p. 26.*
- Producer's surplus:** the difference between the price a seller receives and the lowest price the seller would accept. *Ch. 5, p. 22.*
- Production:** the creation of goods and services that satisfy wants. *Ch. 8, p. 33.*
- Production function:** the relation between the inputs used and the maximum output obtained. *Ch. 8, p. 33.*
- Real wage:** what the money wage can buy. *Ch. 11, p. 46.*
- Returns to scale:** the change in output when all inputs are changed in the same proportion. *Ch. 8, p. 34.*
- Scarcity:** resources are limited while wants are not. *Ch. 1, p. iv.*
- Substitute goods:** goods that can replace each other, such as tea and coffee. *Ch. 3, p. 14.*
- Supply:** the quantity of a good that sellers are willing and able to offer at each price in a given period. *Ch. 6, p. 25.*
- Total utility:** the whole satisfaction from all the units consumed. *Ch. 2, p. viii.*
- Utility:** the satisfaction a person gets from using a good. *Ch. 2, p. viii.*
- Variable cost:** a cost that changes as output changes. *Ch. 9, p. 37.*

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